

FAMILY NAME:
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STUDENT NUMBER:
SIGNATURE:

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

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MATH3311/MATH5335

Mathematical Computing for Finance
Computational Methods for Finance

- (1) TIME ALLOWED – 2 HOURS
- (2) TOTAL NUMBER OF QUESTIONS – 4
- (3) ANSWER ALL QUESTIONS
- (4) THE QUESTIONS ARE OF EQUAL VALUE
- (5) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE
- (6) **ONLY** CALCULATORS WITH AN AFFIXED “UNSW APPROVED” STICKER
MAY BE USED

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [15 marks]

- i) Let $Q \in \mathbb{R}^{n \times n}$.
- a) What property must Q satisfy to be an orthogonal matrix?
 - b) Write down the definition of the matrix norm induced by the Euclidean norm (i.e. 2-norm).
 - c) Show that the 2-norm of Q is 1.
- ii) Let $A \in \mathbb{R}^{n \times n}$.
- a) Write down the QR decomposition of A and state the properties of the matrices.
 - b) Assume you have already computed the QR decomposition of A . Let $b \in \mathbb{R}^n$ be a column vector. State an efficient way of computing

$$b^\top A^\top A b.$$

- c) Assume you have already computed the QR decomposition of A . Assume that $A^\top A$ is invertible. Let $c \in \mathbb{R}^n$ be a column vector. State an efficient way of computing

$$c^\top (A^\top A)^{-1} c.$$

- iii) Given complex numbers $y_0, y_1, \dots, y_{n-1}$ we define the discrete Fourier transform by

$$\hat{y}_k = \sum_{\ell=0}^{n-1} y_\ell e^{2\pi i \ell k / n}, \quad k = 0, 1, 2, \dots, n-1.$$

Prove the inversion formula

$$y_\ell = \frac{1}{n} \sum_{k=0}^{n-1} \hat{y}_k e^{-2\pi i \ell k / n}, \quad \ell = 0, 1, 2, \dots, n-1.$$

Hint: You may assume the formula for the sum of a (complex) geometric progression given by

$$\sum_{k=0}^{n-1} \alpha^k = \frac{1 - \alpha^n}{1 - \alpha}, \quad \text{provided that } \alpha \neq 1.$$

2. [15 marks]

- i) Let $f \in C^3([a, b])$ and $x \in (a, b)$. Consider the expression

$$\frac{3f(a + (h/3)) - 4f(a) + f(a - h)}{h^2} = cf''(a) + O(h). \quad (1)$$

- a) Use Taylor series expansions to find the value of c in (1).
 b) An analyst estimates the second derivative of a function using Formula (1) in MATLAB using double precision, where $a = 2 * 10^6$ and $h = 10^{-12}$. The analyst gets the value 0. Is this correct?
- ii) The Black-Scholes price $P(S, t)$ at time t for a European put option on an asset S with strike price K at maturity T is

$$P(S, t) = N(-d_2)Ke^{-r(T-t)} - N(-d_1)S,$$

where $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-y^2/2) dy$ is the standard normal cumulative distribution function and

$$d_1 = \frac{1}{\sigma\sqrt{T-t}} [\log(S/K) + r(T-t)] + \frac{1}{2}\sigma\sqrt{T-t},$$

$$d_2 = \frac{1}{\sigma\sqrt{T-t}} [\log(S/K) + r(T-t)] - \frac{1}{2}\sigma\sqrt{T-t}.$$

Let p_{mkt} be the market price of the put option with

$$Ke^{-r(T-t)} > p_{mkt} > Ke^{-r(T-t)} - S > 0,$$

and let

$$f(\sigma) = P(S, t) - p_{mkt}.$$

- a) Find $\lim_{\sigma \rightarrow 0^+} f(\sigma)$ and $\lim_{\sigma \rightarrow \infty} f(\sigma)$.
 What does this tell you about the existence of the implied volatility?
- b) **Given** that $SN'(-d_1) = Ke^{-r(T-t)}N'(-d_2)$ (you DO NOT have to prove this equality), where $N'(x) = \frac{1}{\sqrt{2\pi}}\exp(-x^2/2)$ is the standard normal probability density function, show that

$$f'(\sigma) = SN'(-d_1)\sqrt{T-t}.$$

What does this tell you about the uniqueness of the implied volatility?

3. [15 marks]

- i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F , so that

$$\mathbb{P}(Y \leq y) = F(y).$$

- a) Show that the random variable $X = F(Y)$ is uniformly distributed in the interval $[0, 1]$.
- b) How can this connection be used to generate samples of the random variables Y on a computer?
- ii) Transform the expected value

$$\mathbb{E}[h] = \int_{-\infty}^{\infty} h(x)\phi(x) \, dx,$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$ is the standard normal density, into an integral over $[0, 1]$.

- iii) An asset price S_t is assumed to follow a geometric Brownian motion

$$\frac{dS_t}{S_t} = \mu \, dt + \sigma dW_t, \quad t \in (0, T], \quad (2)$$

where W_t is a Wiener process, μ, σ are positive constants and S_0 is given.

- a) What are the key properties of the Wiener process W_t ?
- b) Equation (2) can be discretized using

$$\frac{\Delta S_n}{S_n} = \mu \Delta t + \sigma \sqrt{\Delta t} Z_n,$$

where $Z_0, Z_1, \dots, Z_{N-1}$ are independent, standard normal random variables, $t_n = n\Delta t$, $\Delta t = \frac{T}{m}$, and $\Delta S_n = S_{n+1} - S_n$.

Assume that you can generate independent samples $u_1, u_2, \dots$ which are uniformly distributed in the interval $[0, 1]$. How can you use these samples to simulate the asset prices S_t at times t_n for $n = 0, 1, \dots, m$?

4. [15 marks]

The value $V(S, t)$ of an option on an underlying asset with price S at time t satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV \quad (3)$$

for $0 < S < \infty$ and $0 < t < T$. We assume that the prevailing risk-free interest rate r and the volatility σ are known positive constants.

Consider using (3) to price a put option, which gives the right to sell the underlying asset for the strike price K at expiry T .

- i) Carefully explain what the “initial” conditions are for this problem.
- ii) Carefully explain what the boundary conditions are for this problem.
- iii) Fix a maximum asset price $S_{\max} > 0$ and consider the domain $S \in [0, S_{\max}]$ and $t \in [0, T]$. Set up a grid $(S_j, t_\ell) = (j\Delta S, \ell\Delta t)$ for $j = 0, 1, 2, \dots, n+1$ and $\ell = 0, 1, 2, \dots, m$, where

$$\Delta S = \frac{S_{\max}}{n+1} \quad \text{and} \quad \Delta t = \frac{T}{m}.$$

Let $v_j^\ell \approx V(S_j, t_\ell)$ denote the approximate solution.

- a) At the time point t_ℓ and asset value S_j give
 - A) central difference approximations of $O(\Delta S^2)$ for the partial derivatives

$$\frac{\partial V}{\partial S} \quad \text{and} \quad \frac{\partial^2 V}{\partial S^2};$$

- B) a forward difference approximation of $O(\Delta t)$ for the partial derivative

$$\frac{\partial V}{\partial t}.$$

- b) We want to use an **implicit** finite difference method for computing an approximate solution v_j^ℓ .
 - A) Show that the finite difference equation has the form

$$\alpha_j v_{j-1}^\ell + \beta_j v_j^\ell + \gamma_j v_{j+1}^\ell = v_j^{\ell+1}, \quad (4)$$

and find the coefficients α_j , β_j and γ_j .

- B) What are the discrete “initial” conditions?
 - C) What are the discrete boundary conditions?

END OF EXAMINATION

Q1) i) a) $Q^T Q = I \Leftrightarrow Q^{-1} = Q^T$

b) $\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$

c) $\|Qx\|_2^2 = (Qx)^T Qx = x^T Q^T Qx = x^T I x = x^T x = \|x\|_2^2$

$\Rightarrow \|Q\|_2 = \max_{x \neq 0} \frac{\|Qx\|_2}{\|x\|_2} = \max_{x \neq 0} \frac{\|x\|_2}{\|x\|_2} = 1.$

ii) a) $A = QR$

Q ... orthogonal matrix

R ... upper triangular matrix

b) $A = QR \rightarrow A^T = R^T Q^T \rightarrow A^T A = R^T Q^T Q R = R^T R$

$\rightarrow b^T A^T A b = b^T R^T R b = (Rb)^T Rb$

$\rightarrow$ Compute $y = Rb$
Compute $y^T y$

c) If $A^T A = R^T R$ is invertible $\Rightarrow \det(A^T A) = \det(R^T R) = (\det(R))^2 \neq 0$
 $\Rightarrow \det(R) \neq 0 \Rightarrow R$ invertible; $\Rightarrow A$ invertible

c) $A^{-1} = R^{-1} Q^T \rightarrow (A^T A)^{-1} = R^{-1} R^{-T}$

$\Rightarrow c^T (A^T A)^{-1} c = c^T R^{-1} R^{-T} c = (R^{-T} c)^T (R^{-T} c).$

Let $z = R^{-T} c.$

Step 1: Solve lin. sys. $R^T z = c$ using forward substitution

Step 2: Compute $z^T z.$

$$\text{ii)} \quad \hat{y}_k = \sum_{l=0}^{n-1} y_l e^{2\pi i l k/n}$$

Thus

$$\frac{1}{n} \sum_{k=0}^{n-1} \hat{y}_k e^{-2\pi i \tilde{l} k/n} = \frac{1}{n} \sum_{k=0}^{n-1} \sum_{l=0}^{n-1} y_l e^{-2\pi i \tilde{l} k/n} e^{2\pi i l k/n}$$

$$= \sum_{l=0}^{n-1} y_l \underbrace{\frac{1}{n} \sum_{k=0}^{n-1} e^{2\pi i (l-\tilde{l}) k/n}}_{= \begin{cases} 0 & \text{if } l \neq \tilde{l} \\ 1 & \text{if } l = \tilde{l} \end{cases}} = y_{\tilde{l}}.$$

since if $l = \tilde{l}$ we have $\frac{1}{n} \sum_{k=0}^{n-1} \underbrace{e^{2\pi i (l-\tilde{l}) k/n}}_{=1} = \frac{1}{n} \sum_{k=0}^{n-1} 1 = 1$ and

if $l \neq \tilde{l}$ we have

$$\frac{1}{n} \sum_{k=0}^{n-1} e^{2\pi i (l-\tilde{l}) k/n} = \frac{1 - e^{2\pi i (l-\tilde{l}) n/n}}{1 - e^{2\pi i (l-\tilde{l})/n}} = \frac{1-1}{1 - e^{2\pi i (l-\tilde{l})/n}} = 0.$$

Q2) i) a)

$$f(a+h) = f(a) + \frac{h}{1!} f'(a) + \frac{h^2}{2!} f''(a) + O\left(\frac{h^3}{3!}\right)$$

Hence

$$\frac{3f(a+\frac{h}{3}) - 4f(a) + f(a-h)}{h^2} =$$

$$\begin{aligned} &= \frac{1}{h^2} \left(3f(a) + 3 \frac{h}{3 \cdot 1!} f'(a) + 3 \frac{h^2}{3^2 \cdot 2!} f''(a) + O\left(\frac{h^3}{3^3 \cdot 3!}\right) - 4f(a) \right. \\ &\quad \left. + f(a) - \frac{h}{1!} f'(a) + \frac{h^2}{2!} f''(a) - O\left(\frac{h^3}{3!}\right) \right) \end{aligned}$$

$$= \frac{3}{3^2 \cdot 2!} f''(a) + \frac{1}{2!} f''(a) + O(h) = \left(\frac{1}{6} + \frac{1}{2}\right) f''(a) + O(h)$$

$$= \frac{2}{3} f''(a) + O(h).$$

$$\Rightarrow C = \frac{2}{3}.$$

$$b) a \pm h = 10^6 (2 \pm \underbrace{10^{-18}})$$

< machine epsilon

on computer this number is stored as $2 \cdot 10^6$ (using double precision). Thus

$$\frac{3f(a+\frac{h}{3}) - 4f(a) + f(a-h)}{h^2}$$

is computed as

$$\frac{3f(a) - 4f(a) + f(a)}{h^2} = 0. \quad \text{Answer may be incorrect.}$$

Q2) (i) a)

$$\sigma \rightarrow 0^+ : \text{If } \log \frac{S}{K} + r(T-t) > 0$$

$$\Leftrightarrow \frac{S}{K} e^{r(T-t)} > 1 \Leftrightarrow S > K e^{-r(T-t)}$$

then

$$d_1 \rightarrow \infty, d_2 \rightarrow \infty, N(-d_1), N(-d_2) \rightarrow 0$$

$$\text{If } \log \frac{S}{K} + r(T-t) < 0 \Leftrightarrow S < K e^{-r(T-t)}$$

$$\text{then } d_1, d_2 \rightarrow -\infty, N(-d_1), N(-d_2) \rightarrow 1.$$

Hence, if $S > K e^{-r(T-t)}$ then

$$\lim_{\sigma \rightarrow 0^+} f(\sigma) = -P_{mkt} < 0$$

If $S < K e^{-r(T-t)}$ then

$$\lim_{\sigma \rightarrow 0^+} f(\sigma) = K e^{-r(T-t)} - S - P_{mkt} < 0.$$

$$\sigma \rightarrow \infty: d_1 \rightarrow \infty, d_2 \rightarrow -\infty$$

$$N(d_1) \rightarrow 0, N(-d_2) \rightarrow 1$$

$$\lim_{\sigma \rightarrow \infty} f(\sigma) = K e^{-r(T-t)} - P_{mlt} > 0$$

Since f is continuous in σ and $\lim_{\sigma \rightarrow 0^+} f(\sigma) < 0$ and $\lim_{\sigma \rightarrow \infty} f(\sigma) > 0$
 $\Rightarrow \exists \sigma^*$ s.t. $f(\sigma^*) = 0$. (by intermediate value theorem)

$$\begin{aligned} b) \quad \frac{df}{d\sigma} &= K e^{-r(T-t)} \frac{dN(-d_2)}{d\sigma} - S \frac{dN(-d_1)}{d\sigma} \\ &= K e^{-r(T-t)} N'(-d_2) \left(-\frac{dd_2}{d\sigma} \right) - S N'(-d_1) \left(-\frac{dd_1}{d\sigma} \right) \end{aligned}$$

$$\frac{dd_1}{d\sigma} = -\frac{d_2}{\sigma}, \quad \frac{dd_2}{d\sigma} = -\frac{d_1}{\sigma}$$

$$\begin{aligned} &= K e^{-r(T-t)} N'(-d_2) \frac{d_1}{\sigma} - K e^{-r(T-t)} N'(-d_2) \frac{d_2}{\sigma} \\ &= K e^{-r(T-t)} N'(-d_2) \underbrace{\frac{d_1 - d_2}{\sigma}}_{\sqrt{T-t}} = S N'(-d_1) \sqrt{T-t} > 0. \end{aligned}$$

$$\Rightarrow f'(\sigma) > 0$$

$$\Rightarrow \exists \text{ a unique } \sigma^* \text{ s.t. } f(\sigma^*) = 0.$$

Q3) i)

$$a) P(X \leq x) = P(F(Y) \leq x) = P(Y \leq F^{-1}(x)) = F(F^{-1}(x)) = x \\ \Rightarrow X \text{ is uniformly distributed}$$

b) Generate uniform sample x in $[0,1]$. Set $y = F^{-1}(x)$ to get a sample from Y .

ii) Let $\Phi(y) = \int_{-\infty}^y \phi(x) dx$ and Φ^{-1} its inverse fct.

Set $z = \Phi(x)$, then $\frac{dz}{dx} = \phi(x)$. Then $x = \Phi^{-1}(z)$, $dz = \phi(x) dx$

$$\int_{-\infty}^{\infty} h(x) \phi(x) dx = \int_0^1 h(\Phi^{-1}(z)) dz.$$

iii) a) $W(0) = 0$ a.s.

$\forall t > 0$: $W_{t+u} - W_t$ are independent of W_s , $s < t$.

$$W_{t+u} - W_t \sim N(0, u)$$

W_t is contin. int with prob. 1.

$$b) S_{n+1} = S_n (1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n).$$

Let Φ^{-1} be the inverse of the standard normal cumulative distribution function. Set $Z_n = \Phi^{-1}(x_n)$ for $n=0, 1, \dots, m$.

Compute $S_{n+1} = S_n (1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n)$ for $n=0, 1, 2, \dots, m-1$.

(Q4) i) $V(S, T) = \max\{0, K - S\}$

ii) $V(0, t) = K e^{-r(T-t)}$

$V(S, t) = 0$ as S goes to infinity

iii) a) A)

$$\frac{\partial V}{\partial S}(S_j, t_l) \approx \frac{V_{j+1}^l - V_{j-1}^l}{2 \Delta S}$$

$$\frac{\partial^2 V}{\partial S^2}(S_j, t_l) \approx \frac{V_{j+1}^l - 2V_j^l + V_{j-1}^l}{\Delta S^2}$$

B) $\frac{\partial V}{\partial t}(S_j, t_l) \approx \frac{V_j^{l+1} - V_j^l}{\Delta t}$

b) A)
$$\frac{V_j^{l+1} - V_j^l}{\Delta t} + r S_j \frac{V_{j+1}^l - V_{j-1}^l}{2 \Delta S} + \frac{\sigma^2}{2} S_j^2 \frac{V_{j+1}^l - 2V_j^l + V_{j-1}^l}{\Delta S^2} = r V_j^l$$

$$V_j^{l+1} = V_{j-1}^l \underbrace{\left(\frac{r S_j \Delta t}{2 \Delta S} + \frac{\sigma^2 S_j^2 \Delta t}{2 \Delta S^2} \right)}_{\alpha_j} + V_j^l \underbrace{\left(1 + \frac{\sigma^2 \Delta t S_j^2}{\Delta S^2} + r \Delta t \right)}_{\beta_j} + V_{j+1}^l \underbrace{\left(-\frac{r S_j}{2 \Delta S} - \frac{\sigma^2 S_j^2}{2 \Delta S^2} \right)}_{\gamma_j}$$

B) $V_j^m = \max\{0, K - S_j\}$

C) $V_0^l = K e^{-r(T - \Delta t l)}$, $V_{n+1}^l = 0$